On Digital Post-Distortion Techniques

Ziv Alina and Ofer Amrani

***Abstract*—The potential gain of employing receiver side techniques for mitigating the impact of nonlinear power amplifiers is studied and compared to transmitter side techniques. Receiver side techniques are referred to as Digital Post-Distortion (DPoD). A new DPoD iterative detection scheme, referred to as *Reconstruction Of Distorted Signals* (RODS), is presented. RODS does not attempt to cancel the nonlinear distortion. Rather, it employs the nonlinear components of the received signal to improve detection assuming that the nonlinear model of the power amplifier is known at the receiver. Analytical expressions for the achievable SINR are presented. The advantages of the proposed technique are demonstrated via simulations.**

***Index Terms*—Digital post distortion, digital predistortion, nonlinear equalization, power amplifier nonlinearity cancellation, power amplifier nonlinearity.**

NOMENCLATURE

$x(n)$	n^{th} sample of an OFDM signal
a_k	complex data symbol of the kth subcarrier
N	number of OFDM subcarriers
P	average power of a_k
$\alpha_{2d-1,q}$	PA model: complex coefficient of the $(2d-1)^{th}$ order nonlinearity for the qth delay tap
$2D-1$	maximum nonlinearity order
τ_q	q^{th} delay tap
Q	total number of delay taps
$\tilde{a}_k$	received data symbol at the kth subcarrier
μ	multiplication coefficient of received subcarrier
h_m	channel complex coefficient of mth delay tap
M	total number of channel delay taps
η	nonlinear noise component
σ_{NL}	nonlinear noise power
P_{SAT}	power amplifier saturation power
γ_n	OFDM signal's instantaneous power at the nth sampling interval ($\lvert x(n)^2 \rvert$)
$\underline{\underline{K}}$	NxN matrix relating the nonlinearly distorted subcarriers with the original subcarriers
IBO	Input back-off
CR	Clipping Ratio
BOC	Back-off Correction

I. INTRODUCTION

ONE of the major problems in the analog transmission chain of modern communication systems is Power Amplifier (PA) nonlinearity. Operating in the nonlinear region of a power amplifier results in AM/AM and AM/PM (AM-amplitude; PM-phase) types of distortions of the original signal. It is shown in [1], [2] that for an OFDM signal, the detected subcarriers are comprised of a phase shifted version of the original subcarriers and an inter carrier interference component that can be considered as nonlinear noise. This degrades the achievable Symbol Error Rate (SER) performance as the Input Back-Off (IBO) at the power amplifier input decreases. A second consequence of transmission in the nonlinear region of a power amplifier is that of out-of-band spectral emission mainly due to inter-modulation. It is shown in [3] that for the pth order nonlinearity the, so-called, inter-modulated bands occupy approximately p times the original bandwidth. Since standardization bodies defined strict levels for the allowed adjacent channel interference, out-of-band emission is a legitimate transmitter concern. Nonlinearity is especially severe for high input power. Hence, input signal power is backed-off in an attempt to minimize the phenomena. This degrades power amplifier efficiency, as well as transmission power, and as a consequence, the achievable data throughput in a given channel.

Different model types are used to represent a nonlinear power amplifier. The polynomial model is commonly employed and can account for AM/AM and AM/PM distortions as well as memory effects. Several papers [4]–[6] and [7] discuss these modelling types and their accuracy.

A common approach for linearizing a power amplifier is Digital PreDistortion (DPD), employed at the transmitter side, prior to the power amplifier. DPD attempts to predistort the signal in a fashion that the overall response of the DPD and the power amplifier would be linear. The DPD has to learn the power amplifier response in order to compensate for its distortions. Memory effects in a power amplifier create complications, since the power amplifier response is time dependent. Regularly used in wireless communication, DPD is also found in fiber transmissions [8], [9]. One of the architectures used with DPDs is the indirect learning architecture given in [10]–[12] and [13]. Further discussion of DPD implementation is given in [14] and [15].

A different approach for tackling power amplifier nonlinearity was first suggested by Cioffi *et al.* [16], [17] for OFDM signals. Later named PANC, this approach attempts to estimate the distortion created by the power amplifier, and compensate for it in the receiver. PANC requires knowledge of the power amplifier model's parameters at the receiver side. Gregorio *et al.* [18] considers the feasibility of power amplifier model and channel estimation at the receiver. A further extension was the introduction of joint cancellation of nonlinear distortion for multiple transmitters [19]. This is an acute issue with post-distortion techniques as transmission with a nonlinear power amplifier causes out-of-band emission. Thus a detector, capable of jointly detecting several transmissions, which are interfering

Manuscript received September 12, 2014; revised February 16, 2015 and July 15, 2015; accepted August 19, 2015. Date of publication September 10, 2015; date of current version December 18, 2015. The associate editor coordinating review of this manuscript and approving it for publication was Prof. Ravi Raich.

The authors are with the School of Electrical Engineering, Tel Aviv University, Ramat Aviv 69978, Tel-Aviv, Israel (e-mail: ofera@eng.tau.ac.il; elinaziv@post.tau.ac.il).

Color versions of one or more of the figures in this paper are available online at http://ieeexplore.ieee.org.

Digital Object Identifier 10.1109/TSP.2015.2477806

1053-587X © 2015 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See http://www.ieee.org/publications_standards/publications/rights/index.html for more information.
Authorized licensed use limited to: CAMBRIDGE UNIV. Downloaded on July 08,2026 at 03:40:52 UTC from IEEE Xplore. Restrictions apply.

with each other, is a necessity. Extension of the concept of PANC to SC-OFDM was also studied in [20] and [21].

This work attempts to further understand the advantages of post-distortion techniques. We also present a new post-distortion algorithm, distinct from PANC by the utilization of the distorting terms in the nonlinear signal. The proposed algorithm can provide improvement in performance over DPD and PANC techniques. We also upper bound the achievable Signal to Interference-plus-Noise Ratio (SINR) with a nonlinear power amplifier.

The rest of the paper is organized as follows. Section II introduces the power amplifier polynomial model, the OFDM signal model and an expression for the distorted signal received by the detector. A review of present predistortion and post-distortion techniques is given in Section III. Also shown in this section is the potential gain in post-distortion techniques over predistortion. Provided in Section IV is the new iterative detection algorithm with formulation of theoretical SINR along with SINR value for PANC. Section V compares the performance of the proposed algorithm with PANC and DPD algorithms via simulations. Conclusion are drawn in Section VI.

II. Background

A. Power Amplifier Model

The general model for describing nonlinear behaviour is the well known Volterra series [22]. A simplified model, commonly used to describe a power amplifier is the following (discrete-time) memory polynomial model:

$$y(n) = \sum_{d=1}^{D} \sum_{q=0}^{Q} \alpha_{2d-1,q} x(n-\tau_q) |x(n-\tau_q)|^{2(d-1)}, \quad (1)$$

where x and y are, respectively, the input and output of the power amplifier; $\alpha_{2d-1,q}$ is the complex coefficient of the $2d-1th$ order nonlinearity for the qth delay tap; τ_q is the qth delay tap and Q is the total number of delay taps. Note that if the signal guard interval (cyclic prefix) is greater than the memory, τ_Q, of the power amplifier then each OFDM symbol is not interfered by other symbols. The above model differs from the Volterra series in that it does not consist of cross terms between taps.

For the memoryless case where $Q = 0$ and $\tau_q = 0$, (1) reduces to

$$y(n) = \sum_{d=1}^{D} \alpha_{2d-1} x(n) |x(n)|^{2(d-1)}. \quad (2)$$

For simplicity of the analysis we initially focus on the memoryless scenario. An extension of the algorithm derived in Section IV for dealing with memory polynomial models is given in Appendix C.

B. Signal Model

The complex baseband discrete representation of an OFDM signal of length N with the supplement of a Cyclic Prefix (CP) is given by

$$x(n) = \frac{1}{N} \sum_{k=0}^{N-1} a_k e^{j2\pi kn/N}; \quad -N_g \le n \le N-1, \quad (3)$$

where k is the subcarrier index; $\{a_k\}$ are the complex data symbols, which are assumed to be independent, identically distributed with zero mean and variance P; n is the discrete time domain index and N_g is the index associated with the guard interval. By the Central Limit Theory when N is large, $x(n)$ is assumed to be a complex Gaussian process with zero mean and variance $P_{av} = \frac{P}{N}$. The complex data symbol at the k'th subcarrier obtained after removal of the guard interval and DFT at the receiver side is

$$\tilde{a}_k = \sum_{n=0}^{N-1} y(n) e^{-j2\pi kn/N} + n_0(k). \quad (4)$$

Where $n_0(k)$ is the AWGN. Substituting (2) into (4) yields

$$\tilde{a}_k = \sum_{n=0}^{N-1} \sum_{d=1}^{D} \alpha_{2d-1} x(n) |x(n)|^{2(d-1)} e^{-j2\pi kn/N} + n_0(k). \quad (5)$$

Bussgang Theorem [23] states that the output of a nonlinear operation on a Gaussian signal is composed of a scaled version of the input and an uncorrelated noise term. Following this theorem Bohara showed [1], [2], that $\tilde{a_k}$ can be expressed as

$$\tilde{a_k} = a_k \mu + \eta + n_0(k), \quad (6)$$

where a_k, the original symbol transmitted at the k'th subcarrier, is multiplied by a complex factor μ; and η is referred to as the nonlinear noise component and is a (nonlinear) deterministic function of all N subcarriers. Following the derivation of η in [1], it can be shown that μ and η are uncorrelated. For a memoryless power amplifier μ and η are constant for all subcarriers. An explicit derivation of μ for the memoryless case is given in Appendix B.

Based on (6) the SINR for each subcarrier can be expressed as

$$\text{SINR} = \frac{P|\mu|^2}{\sigma^2_{\text{NL}} + N_0} \quad (7)$$

where N_0 is the AWGN spectral density. We note that (7) quantifies the amount of interference experienced by each subcarrier due to the nonlinear noise and the AWGN.

The nonlinear noise experienced by each subcarrier may admit an elegant closed form expression. For example, η associated with a third degree memoryless power amplifier has the following variance [2]:

$$\sigma^2_{\text{NL}} = \frac{|\alpha_3|^2 P^3 (3N^2 - 11N + 12)}{N^4}. \quad (8)$$

III. Digital Linearization Techniques

A. Digital Predistortion

Digital PreDistortion (DPD) is the most commonly used solution in modern day communication for mitigating the influence of nonlinearity in power amplifiers. It follows from (2) that the linear term in a memory polynomial model is $\alpha_{1,0} x(n)$. The remaining terms typically suppress the linearity of the power amplifier, which altogether result in the dotted blue curve shown

Authorized licensed use limited to: CAMBRIDGE UNIV. Downloaded on July 08,2026 at 03:40:52 UTC from IEEE Xplore. Restrictions apply.

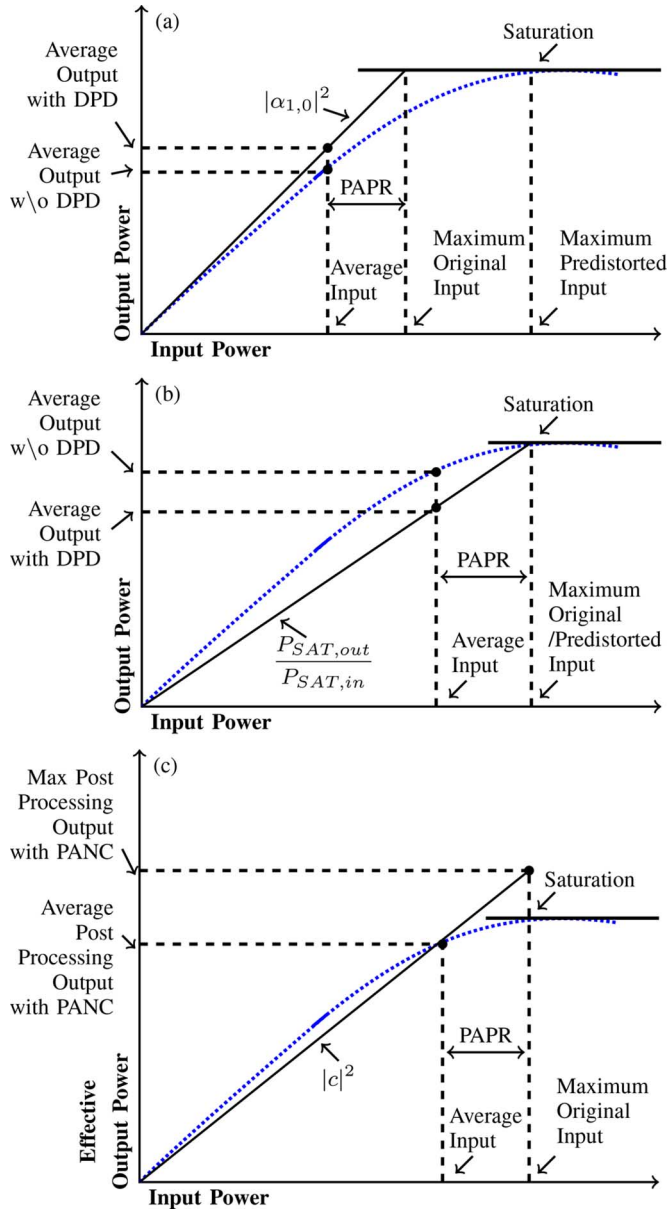


Fig. 1. Effective gain curves for different linearization techniques; dotted blue curve represents the nonlinear amplifier and is the same for all subfigures. (a) Response with DPD linearized to maximum gain (b) Response with DPD linearized for gain of the maximum input power level (c) PANC post processing gain.

in Fig. 1. Hence, a linear power amplifier would be composed solely of $\alpha_{1,0}x(n)$.

Fig. 1 presents the effective gain curves for different linearization techniques. The DPD of Fig. 1(a), henceforth denoted as DPD(a), linearizes the power amplifier curve so as to obtain a gain of $\alpha_{1,0}$. Limited by power supply and power amplifier efficiency, a power amplifier cannot practically exceed the (output) saturation level. Hence the maximum allowed power at the input of DPD(a) is smaller than $P_{\text{in}}(\text{SAT})$, the input level that brings the output to saturation, as shown in Fig. 1(a).

The DPD of Fig. 1(b), DPD(b), is set to provide smaller gain and reaches the output saturation level for the maximum possible input power. Note that both DPD(a) and DPD(b) provide the same average and maximum output powers, as would any DPD whose gain is selected in the range defined by the two DPDs above. These DPDs are therefore equivalent in terms of receiver side SINR. Clearly, when assuming perfect linearization, out-of-band emission is eliminated, and the effective DPD and power amplifier response is similar to that of a limiter (with the maximum output power clipped to the saturation level).

TABLE I
POWER AMPLIFIER NONLINEARITY CANCELLATION—ALGORITHM

Given the received symbols, $\underline{\tilde{a}}$, and the coefficient μ, explicitly defined in (23), initial estimation ($\underline{\hat{a}}^{(0)}$) of the transmitted subcarriers is given by:

$$\underline{\hat{a}}^{(0)} = \frac{\underline{\tilde{a}}}{\mu}$$

m^{th} estimation ($\underline{\hat{a}}^{(m)}$) of transmitted subcarriers:

- Estimate the time domain signal:

$$\underline{x}^{(m)} = IFFT(\underline{\hat{a}}^{(m-1)})$$

- Estimate distortion:

$$\underline{d}^{(m)} = y(\underline{x}^{(m)}) - c \cdot \underline{x}^{(m)}$$

- Estimate distortion in frequency domain:

$$\underline{D}^{(m)} = FFT(\underline{d}^{(m)})$$

- Subtract distortion and estimate $\underline{\hat{a}}^{(m)}$:

$$\underline{\hat{a}}^{(m)} = \frac{\underline{\tilde{a}} - \underline{D}^{(m)}}{c}$$

B. Digital Post-Distortion

Suppose now that the task of mitigating power amplifier nonlinearity is handed over to the receiver—to be termed Digital Post-Distortion (DPoD). An indication of the potential gain attainable with such an approach is evident form Fig. 1(b): the average output power when employing DPD is lower than without it.

An immediate downside of receiver-side post distortion may be out-of-band emission (spectral regrowth) due to power amplifier nonlinearity. The allowed level of such emission is well defined by wireless communication standards. The 3GPP LTE standard defines a minimum drop of 35 dB in power spectral density (PSD) at the channel edge. The current 802.11 WIFI standard is less stringnet requiring but 20 dB drop in the PSD at the channel edge, thus rendering this communication scheme more suitable for DPoD implementation. Appendix E provides an example of an OFDM signal amplified by one of the nonlinear amplifiers given in Table III, that obeys the 802.11ac spectral mask requirements. Other communication technologies, less susceptible to out-of-band emission, such as wire-line and optical communications, can gain from DPoD-based receiver architectures.

1) Power Amplifier Nonlinearity Cancellation: Cioffi *et al.* [16], [17] presented a DPoD algorithm for OFDM signals known as Power Amplifier Nonlinearity Cancellation (PANC). For a power amplifier whose behavior is modeled by (2), the PANC algorithm steps are briefly summarized in Table I.

PANC, as implied by its name, is an algorithm whose aim is to estimate and cancel the linearity-suppressing terms in (2). By

Authorized licensed use limited to: CAMBRIDGE UNIV. Downloaded on July 08,2026 at 03:40:52 UTC from IEEE Xplore. Restrictions apply.

doing so, PANC may theoretically approach (when the distortion is perfectly estimated) $y(n) = \alpha_{1,0} \cdot x(n)$, thus effectively realizing the available gain as obtained by DPD(a) jointly with an extended input dynamic-range as DPD(b). However, Bussgang Theorem suggests that the nonlinear distortion is minimized when using the model $y(n) = \mu \cdot x(n) + d(n)$ (which is the time domain representation of (6)). In the squeal we demonstrate the gain achieved by setting the complex constant c to $c = \mu$, as opposed to using $c = \alpha_{1,0}$. We use $c = \mu$, supported by Bussgang Theorem, to quantify the improvement in performance of PANC over DPD. The post processing power gain and SINR achieved by PANC are, respectively, $|\mu|^2$ and

$$\mathrm{SINR}_{\mathrm{PANC}} = \frac{P \cdot |\mu|^2}{N_0}. \tag{9}$$

This is demonstrated in Fig. 1(c), where the effective power, as viewed by a PANC-based receiver, is plotted as a function of the input power. Compare (9) to the initial SINR associated with the received subcarriers given by (7). The SNR for a power amplifier perfectly linearized by a DPD is deducted by replacing μ with the gradient of the linear curve shown in Fig. 1(b)

$$\mathrm{SNR}_{\mathrm{DPD}} = \frac{P \cdot \frac{P_{\mathrm{SAT,out}}}{P_{\mathrm{SAT,in}}}}{N_0}. \tag{10}$$

By subtracting the nonlinear distortion terms, PANC may be viewed as an iterative "zero-forcing" detection scheme. This scheme coincides with the Maximum Likelihood (ML) solution only when the distortion is perfectly estimated. Cioffi [16] hence referred to PANC as *quasi-ML* detection.

A superior, yet more complex, algorithm based on a steepest descent search is given in [24]. An improved version of PANC, proposed by Haimovich *et al.* [25], utilizes coding for the estimation of the distortion terms in a similar fashion to Turbo equalization. The decoder output is used to provide a better estimation of the transmitted subcarriers, with which the distortion terms are more reliably estimated. Since these modifications to PANC can be similarly incorporated in the framework of the RODS algorithm presented in Section IV, we shall employ the original PANC algorithm (described in Table I) as a DPoD method of reference to RODS.

2) Multi-User Power Amplifier Nonlinearity Cancellation: The improved post processing SINR of PANC compared to DPD motivates the use of receiver side compensation for power amplifier nonlinearity. Because DPD is carried out in the digital domain at the transmitter, it is capable of mitigating out-of-band emission regardless of the carrier frequency of the transmitted signal. This is critical for many applications, such as cellular networks with dynamic frequency allocations for user endpoints. Without DPD, the emission of power outside the specific frequency allocation of a user cannot be mitigated by an analog filter which follows the power amplifier, as it is unrealistic to implement frequency-adjustable filters having the required sharpness. This problem was tackled by Gregorio [19] who considered joint cancellation of power amplifier nonlinearity for multiple transmitters (or multi-user PANC). A later version [20] targeted SC-OFDM systems. In a multi-user system a frequency band is divided, not necessary evenly, among several independent transmitters. At the receiver side, the PANC algorithm is performed across the entire band, with a single receiver iteratively detecting the data symbols sent from each user. With each iteration the distortion created by a particular user is estimated and subtracted across the band from the received subcarriers to provide better detection of all other users. For DPoD in general, and PANC in particular, joint cancellation of power amplifier nonlinearities allows multiple users to transmit in a given band without having to linearize their power amplifiers. It is only required that each transmitter filters out out-of-band emission using a (fixed-frequency) analog filter. With proper standardization, the notion of multi-user post distortion may be suitable for incorporation in future releases of WIFI or 5G.

IV. Detection of Nonlinear OFDM Signals—The RODS Algorithm

In this section we take on a different approach to post-distortion and present a new DPoD algorithm. Unlike PANC, we detect the "nonlinear" received signal without canceling the nonlinear estimated distortion. As will be shown in the sequel, this approach outperforms PANC SER-wise. We start by substituting (3), the IDFT representation of $x(n)$, into (5) in order to derive an expression for the received subcarriers:

$$\begin{aligned}\tilde{a}_k = {} & \alpha_1 \cdot a_k \\ & + \frac{1}{N} \sum_{d=2}^{D} \alpha_{2d-1} \sum_{n=0}^{N-1} \sum_{p=0}^{N-1} a_p \gamma_n^{d-1} e^{j2\pi n(-k+p)/N} \\ & + n_0(k),\end{aligned} \tag{11}$$

where $\gamma_n = |x(n)|^2$. After some manipulation, detailed in Appendix A, one can obtain a simple expression for relating the distorted data symbols with the original ones:

$$\underline{\tilde{a}} = \underline{\underline{K}} \cdot \underline{a} + \underline{n_0}, \tag{12}$$

where the matrix $\underline{\underline{K}}$ is dependent on the power amplifier coefficients $\{\alpha_{2d-1}\}$ and on the instantaneous signal power γ_n. $\underline{n_0}$ is an AWGN noise vector. The linear representation of $\tilde{a}$ (12) calls for the following Least Squares (LS) solution for estimating the original data symbols:

$$\underline{\hat{a}} = (\underline{\underline{K}}^* \cdot \underline{\underline{K}})^{-1} \cdot \underline{\underline{K}}^* \cdot \underline{\tilde{a}}, \tag{13}$$

with $\underline{\hat{a}}$ being the estimation of the transmitted data symbols, $\underline{a}$, based on the received symbols $\underline{\tilde{a}}$.

While the polynomial coefficients associated with the power amplifier are assumed to be known at the receiver side, the instantaneous power of the signal, γ_n, is unknown, and needs to be estimated in order to derive the matrix $\underline{\underline{K}}$. The proposed detection scheme, to which we shall henceforth refer as *Reconstruction Of Distorted Signals* (RODS), iteratively estimates $\underline{\underline{K}}$ and uses (13) to extract an estimation of the transmitted subcarriers. RODS algorithm is detailed in Table II and shown as a block diagram in Fig. 2. Perfect knowledge of $\underline{\underline{K}}$ allows for perfect linearization of the model given by (1) via (12). In this case, the LS solution (13) is also the ML solution for (12). Thus RODS iteratively approaches the ML solution of (1), hence realizes the dotted blue curve of Fig. 1. Given $\underline{\underline{K}}$ the variance of the estimation error associated with the kth subcarrier is given by

$$Var(\hat{a_k}) = N_0 \cdot (\underline{\underline{K}}^* \cdot \underline{\underline{K}})^{-1}_{kk}, \tag{14}$$

Authorized licensed use limited to: CAMBRIDGE UNIV. Downloaded on July 08,2026 at 03:40:52 UTC from IEEE Xplore. Restrictions apply.

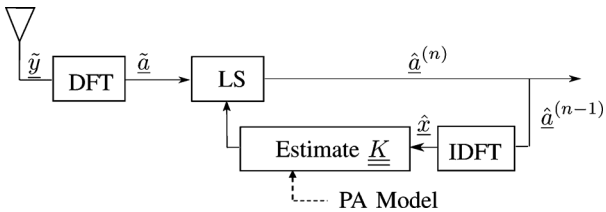


Fig. 2. Reconstruction Of Distorted Signals—Block Diagram.

TABLE II
RECONSTRUCTION OF DISTORTED SIGNALS—ALGORITHM

Given the received symbols, $\underline{\tilde{a}}$, and the coefficient μ, explicitly defined in (23), initial estimation ($\underline{\hat{a}}^{(0)}$) of the transmitted subcarriers:

$$\underline{\hat{a}}^{(0)} = \frac{\underline{\tilde{a}}}{\mu}$$

m^{th} estimation ($\underline{\hat{a}}^{(m)}$) of transmitted subcarriers:

- Calculate the time domain signal:

$$\underline{x}^{(m)} = IFFT(\underline{\hat{a}}^{(m-1)})$$

- Calculate the signal's instantaneous power:

$$\underline{\gamma}^{(m)^T} = \left(|x_0^{(m)}| \dots |x_{N-1}^{(m)}|\right)$$

- Using $\underline{\gamma}^{(m)}$ estimate $\underline{\underline{K}}^{(m)}$ as detailed in Appx. A
- Least Squares solution for $\underline{\hat{a}}^{(m)}$:

$$\underline{\hat{a}}^{(m)} = (\underline{\underline{K}}^{*^{(m)}} \cdot \underline{\underline{K}}^{(m)})^{-1} \cdot \underline{\underline{K}}^{*^{(m)}} \cdot \underline{\tilde{a}}$$

and the mean error variance is $N_0 \cdot E\{\mathrm{diag}([\underline{\underline{K}}^* \cdot \underline{\underline{K}}]^{-1})\}$. Simulations reveal that the mean error variance is approximately $\frac{N_0}{|\mu|^2}$. Thus, RODS converges to the following SINR

$$\mathrm{SINR}_{\mathrm{RODS}} = \frac{P \cdot |\mu|^2}{N_0}. \tag{15}$$

Note that the complex coefficient μ is constant only in the case of a memoryless power amplifier. For memory polynomial model μ varies among subcarriers [1], hence the mean SINR would be a function of $E[|\mu(k)|^2]$. Equations (9) and (15) suggest that PANC and RODS converge to post processing gain of $|\mu|^2$. Section V supports the above SINR behavior by simulations. Nevertheless, as will also be shown in the next section, RODS provides superior performance in terms of symbol error rate.

A. Multiple Input Scenario

In Section III-B2 we discussed the concept of joint post distortion processing of multiple transmissions (i.e., users) as an approach for coping with the out-of-band emission caused by nonlinear power amplifiers. This idea closely relates to the concept of Multi User Multiple Input Multiple Output (MU-MIMO). In a cellular MU-MIMO system the base station allocates users overlapping frequency resources. The users to participate in this transmission scheme are chosen so that their combined channel matrix is invertible, allowing the processing at the receiver to separate the transmissions. This concept may be coupled with DPoD. Assume a MU-MIMO system where N users are allocated overlapping resources, and an equalizer in the receiver outputs N separate streams, each corresponding to the one of the users. DPoD would then have to process each of the streams independently, thus complexity would be linear with the number of power amplifiers. Note, however, that there could be MIMO scenarios where DPoD complexity is not necessarily linear in the number of power amplifiers, for example due to precoding.

TABLE III
POWER AMPLIFIER MODELS USED IN SIMULATION

Model	Coefficients	Delay $[T_S]$
PA(1)	$\alpha_1 = 3.6181,\quad \alpha_3 = 1.2505,\quad \alpha_5 = -1.3599,$ $\alpha_7 = 0.53232,\quad \alpha_9 = -0.11166,\quad \alpha_{11} = 0.012985$ $\alpha_{13} = -0.00078,\quad \alpha_{15} = 0.000018687$	$\tau_0 = 0$
PA(2)	$\alpha_1 = 0.9798 - 0.2887i,\quad \alpha_3 = -0.2901 + 0.435i$	$\tau_0 = 0$
PA(3)	$\alpha_{1,0} = 0.9800 - 0.3000i,\quad \alpha_{3,0} = -0.3000 + 0.4200i$	$\tau_0 = 0$
	$\alpha_{1,1} = 0.0600 + 0.0300i,\quad \alpha_{3,1} = -0.0200 + 0.0500i$	$\tau_1 = 1$
	$\alpha_{1,2} = -0.0100 + 0.0200i,\quad \alpha_{3,2} = 0.0200 - 0.0100i$	$\tau_2 = 5$
	$\alpha_{1,3} = 0.0200 + 0.0800i,\quad \alpha_{3,3} = -0.0100 - 0.0800i$	$\tau_3 = 10$
PA(4)	$\alpha_1 = 14.8526 - 0.1337i,\quad \alpha_3 = -23.1899 + 6.9785i$ $\alpha_5 = 30.5226 - 1.9699i,\quad \alpha_7 = -21.5517 - 4.7097i$ $\alpha_9 = 6.0311 + 2.7527i$	$\tau_0 = 0$

V. SIMULATION RESULTS

The discussed techniques are evaluated in this section for several different amplifiers whose parameters are provided in Table III. PA(1) is memoryless, weakly nonlinear and exhibits AM/AM distortion only. PA(2) is memoryless and exhibits moderate nonlinearity along with AM/PM distortion. PA(3) is the memory polynomial version of PA(2). PA(4) is a second nonlinear memoryless power amplifier with AM/PM distortion. For the simulations we assumed an OFDM signal with 256 subcarriers modulated by 64QAM or QPSK. The average PAPR of a 64QAM OFDM signal is $\sim$ 7.8 dB. Note that the receiver sample rate was set to twice the Nyquist rate. Filtering was incorporated in the receiver followed by down-sampling to Nyquist rate. This insured that the majority of the out-of-band emission due to power amplifier nonlinearity did not alias in-band. A scenario of sampling at Nyquist rate was also simulated (not shown in the paper). Results show similar gaps in performance between DPD and DPoD methods. This may be of interest for systems where there is a requirement for sampling at Nyquist rate and where there are no adjacent channels that would alias in-band.

A. SINR Performance

Table IV presents the received SINR for the RODS DPoD technique as a function of the so-called, Input Back-Off (IBO). Input back-off is defined as

$$\Upsilon = \frac{P_{in}(\mathrm{SAT})}{P_{\mathrm{in}}(\mathrm{avg})}, \tag{16}$$

where $P_{\mathrm{in}}(\mathrm{avg})$ is the average power of the signal at the power amplifier input. Results are provided for two cases—PA(2) and PA(3), the memoryless and memory polynomials versions of the

Authorized licensed use limited to: CAMBRIDGE UNIV. Downloaded on July 08,2026 at 03:40:52 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
RODS: EMPIRICAL AND THEORETICAL SINR [DB]

	IBO, Υ [dB]	9	7	5
PA(2)	DPD	24.8	26.3	24.1
	Iteration 0	20.5	17.5	12
	RODS Theoretical	27.5	29.3	30.9
	RODS Empirical	27.5	29.3	30.7
PA(3)	RODS Empirical	27.5	29.2	30.4

same power amplifier. We present the "empirical" SINR obtained via simulations (after 4 iterations) along with the theoretical SINR given by (15). As reference for PA(2), we include the empirical SINR obtained for ideal DPD, i.e., assuming that the power amplifier is perfectly linearized. The AWGN variance used in the simulations was set 30 dB bellow a power of $|\alpha_1|^2 \cdot P_{\text{in}}(\Upsilon = 7\,\text{dB})$.

Several observations can be made with respect to the results summarized in Table IV:

1) The theoretical SINR results closely match the empirical ones for non-heavily clipped signals;
2) As will be demonstrated in the next subsection, due to the bursty nature of the symbol errors, inconsistency occurs between SINR and symbol error rate. In particular, error floor is observed for $\Upsilon < \text{PAPR}$. This is not reflected by the SINR, empirical or theoretical, as seen in an input back-off of 5 dB.

B. PA(1): AM/AM Distortion

Hereinafter we study the symbol error rate performance of RODS, PANC and DPD for different input back-off levels. Let us denote by $SNR^{\star}$ the ratio between a constant power of $|\alpha_1|^2 \cdot P_{\text{in}}(\Upsilon = 7\,\text{dB})$ and the AWGN variance.

Fig. 3 presents the simulation results of the aforementioned techniques applied to PA(1). The two variants of the PANC algorithm detailed in Section III-B1 are studied. The first variant, termed PANC_{α}, employs $\alpha_{1,0}$, the linear term of the power amplifier model, for estimating the nonlinear distortion. The second variant, PANC_{μ}, employs μ for the estimation, where μ is known to minimize the distortion. In the scenario given in Fig. 3 the two variants yield similar performance, hence they were commonly referred to as PANC.

Note that RODS and PANC exhibit similar behavior. Compared to ideal DPD, DPoD techniques achieve significant improvement in performance of 2.5 dB.

Note that performance flaring occurs at input back-off of 8 dB. This can be attributed to signal realizations whose PAPR exceed the saturation level of the power amplifier, thus resulting with increased error probability.

In order to establish this assertion, let us first define the (natural) clipping ratio for a signal as

$$\text{CR} = \frac{max(P_{\text{in}})}{P_{\text{in}}(\text{SAT})}, \tag{17}$$

where $max(P_{\text{in}})$ is the maximal instantaneous power of the signal. We shall limit the allowed clipping ratio of the signal to 3 dB (twice the input saturation level of the power amplifier);

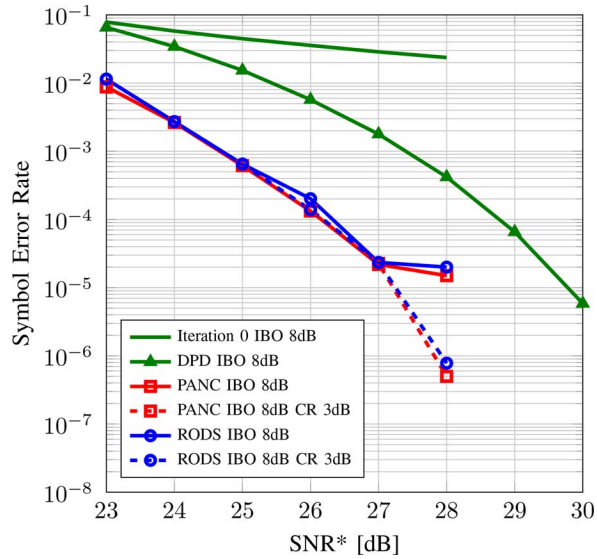


Fig. 3. PA(1), Symbol error rates. RODS (circle marker) 3 iterations, PANC (square marker) 5 iterations and DPD (triangle marker): N = 256, 64QAM, AWGN channel.

signal realizations exceeding this threshold will be discarded by the transmitter.

Simulations reveal that for input back-off of 8 dB, less than 0.1% of the OFDM symbols need to be discarded. Also, the relevant symbol error-rate curve, labeled *CR* in Fig. 3, shows that the error flaring is completely eliminated.

C. Back-Off Correction

Since discarding high PAPR OFDM symbols is typically not a viable option, we instead introduce the so-called *Back-off Correction* (BOC). According to back-off correction, any symbol whose PAPR exceeds a predefined clipping ratio threshold, is further backed-off on top of the preset input back-off. Back-off correction is particularly beneficial for highly non linear power amplifiers.

D. Low SNR Regime

For studying the performance of post distortion techniques in the low SNR regime we employ PA(2). We simulate an OFDM signal with 256 subcarriers modulated by QPSK. Since QPSK is less susceptible than 64QAM to noise and nonlinear distortion, the input back-off of choice (i.e., the back-off that yields minimum symbol error rate) is lower than 64QAM.

Fig. 4 provides the performance for the QPSK scenario. Note that RODS outperforms PANC_{μ} and DPD by some 2 dB. Also, the gap between the two variants of PANC is more pronounced here, the reason for which is that at low input back-off the coefficients μ and $\alpha_{1,0}$ are quit different. This behavior is also well demonstrated by simulation results of PA(4).

E. Memory Effects

Figs. 5 and 6 provide the symbol error rate performance for the amplifiers PA(2) and PA(3). Note that the input back-off for DPD must be set to 8 dB, otherwise error floor occurs; for PANC input back-off of 9 dB yields the best results. We draw the following conclusions:

Authorized licensed use limited to: CAMBRIDGE UNIV. Downloaded on July 08,2026 at 03:40:52 UTC from IEEE Xplore. Restrictions apply.

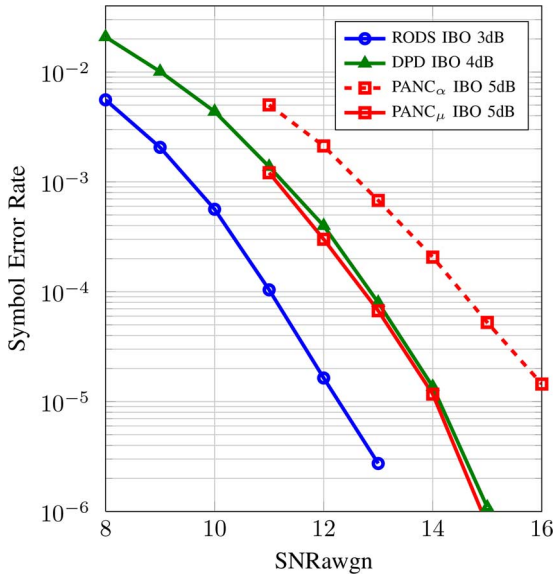


Fig. 4. QPSK with PA(2), Symbol error rates. RODS (circle marker) 3 iterations, PANC (square marker) 5 iterations and DPD (triangle marker): N = 256, QPSK, AWGN channel.

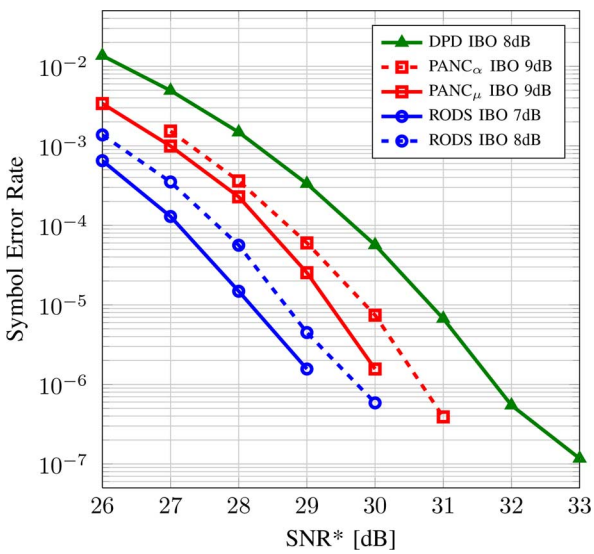


Fig. 5. PA(2), Symbol error rates. RODS (circle marker) 4 iterations, PANC (square marker) 5 iterations and DPD (triangle marker): N = 256, 64QAM, AWGN channel.

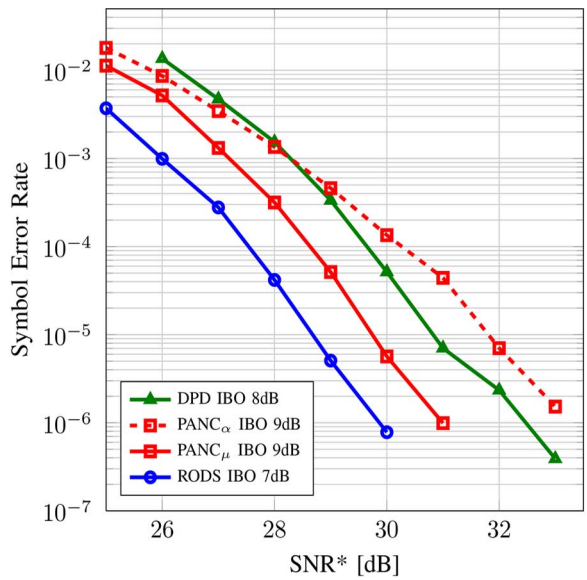


Fig. 6. PA(3), Symbol error rates. RODS (circle marker) 4 iterations, PANC (square marker) 5 iterations and DPD (triangle marker): N = 256, 64QAM, AWGN channel.

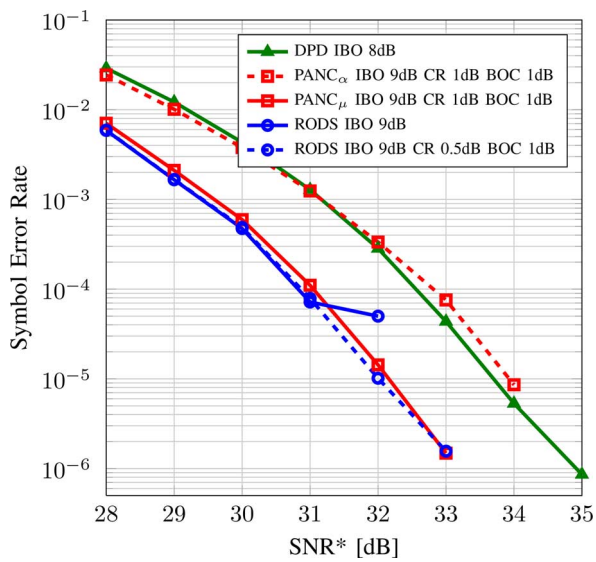


Fig. 7. PA(4), Symbol error rates. RODS (circle marker) 3 iterations, PANC (square marker) 5 iterations and DPD (triangle marker): N = 256, 64QAM, AWGN channel.

1) RODS outperforms PANC by approximately 1 dB for PA(2) and PA(3). The improvement with respect to an ideal DPD is 2.5 dB.
2) Degradation of some 0.5 dB is observed when the power amplifier is modeled as memory polynomial (PA(3)) as compared to a memoryless power amplifier (PA(2)). This validates the suitability of DPoD techniques for power amplifiers with memory.
3) PANC$_\alpha$, on the other hand, exhibits severe degradation of 2 dB due to memory, and is inferior even to the ideal DPD. Recall that μ is frequency dependent for PA(3) ((34)). This spectral dependency is ignored by PANC$_\alpha$, which can explain its severe degradation.

F. PA(4): AM/PM Distortion

Fig. 7 provides symbol error rate performance for PA(4). For this nonlinear power amplifier, at SER $= 10^{-5}$, RODS and PANC$_\mu$ outperform DPD by 2 dB. Notice that with proper setting of the *Back-off Correction* and *Clipping Ratio* threshold, the vast majority of error bursts caused by high-PAPR signals are eliminated without any loss in throughput.

The two variants of PANC exhibit significant gap in performance. To explain this behavior we argue as follows: A practical power amplifier typically has 3, quite distinct, operating regions: (close to) linear region; *strong nonlinearity region*; and

Authorized licensed use limited to: CAMBRIDGE UNIV. Downloaded on July 08,2026 at 03:40:52 UTC from IEEE Xplore. Restrictions apply.

non one-to-one mapping, or *compression*, region. Non of the digital distortion techniques, RODS, PANC or DPD, target the compression region. It is the *strong nonlinearity region* that greatly impacts the PANC_α behavior, and consequently degrades its performance. In this region the power amplifier transmission function is dominated by nonlinear terms, rather than the linear first-order term. Here, the convergence of PANC_α to a linear power amplifier model degrades. This observation seems to suggest a limit on the effective usage of PANC_α to fairly linear memoryless power amplifiers.

While RODS and PANC_μ perform successfully in the *strong nonlinearity region*, the number of iterations required by RODS is smaller. Therefore RODS may be superior as far as latency is concerned.

G. Multipath Channel

A practical communication use case is likely to include a fading channel. It is therefore informative to compare the performance of the above linearization techniques for a multipath channel.

At the receiver side an equalizer can be used to flatten the channel response. Assuming perfect equalization, the equalized signal can be modeled as flat faded perturbed by additive colored Gaussian noise. Alternatively, the distorted signal can be modeled by combining the power amplifier model given in (1) with the channel impulse response to yield the following memory polynomial model

$$y(n)=\sum_{d=1}^{D}\sum_{q=0}^{Q}\sum_{m=0}^{M}h_m\alpha_{2d-1,q}x(n-\tau_q-B_m)\cdot|x(n-\tau_q-B_m)|^{2(d-1)}, \quad (18)$$

where M is the total number of delay taps, h_m is the channel complex coefficient of the mth delay tap and B_m is the mth delay value. Note that if the signal cyclic prefix is greater than the memory length of the channel and power amplifier combined, τ_Q+B_M, then each OFDM symbol is not interfered by other symbols.

In the simulations we assume perfect channel and power amplifier knowledge at the receiver side (a discussion of methods for power amplifier model and channel estimation is given in Appendix D). We simulate the memoryless PA(2) with a multipath channel given by the following channel impulse response

$$h(m)=0.936\delta(m)+(0.2093+0.2093i)\delta(m-4)-0.1664i\delta(m-15)+(-0.0662+0.0662i)\delta(m-30). \quad (19)$$

The RODS algorithm is readily applicable for the scenario defined by (18). For DPD the following equalization is performed prior to detection:

$$(\underline{H}^*\underline{H})^{-1}\cdot\underline{H}^*\cdot\underline{\tilde{Y}} \quad (20)$$

Results are given in Fig. 8. Note that RODS outperforms PANC and DPD. Moreover, the gains observed in the AWGN scenario, as depicted in Fig. 5, are preserved also under the multipath channel scenario.

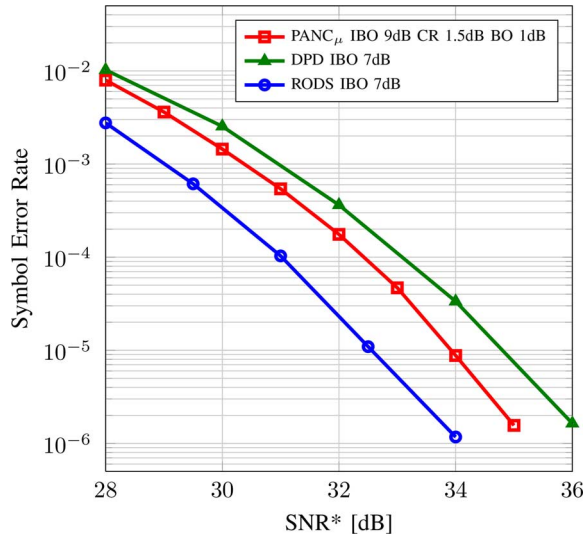


Fig. 8. Multipath Channel with PA(2), Symbol error rates. RODS (circle marker) 4 iterations, PANC (square marker) 5 iterations and DPD (triangle marker): N = 256, 64QAM.

VI. Conclusion

We studied the potential gain of digital post-distortion over predistortion techniques for linearizing a power amplifier whose transmission model is nonlinear. A new iterative DPoD detection scheme, termed RODS, was developed and it's theoretical SINR was derived. We combined *Clipping Ratio* threshold with *Back-off Correction* as means for mitigating sporadic errors, which in turn allowed further decrease in the input back-off. Comparison of the proposed RODS scheme with ideal DPD and PANC was conducted via simulations, revealing RODS to provide the best performance. In some cases, e.g., PA(1) and PA(4), PANC_μ achieves similar performance to RODS, as suggested by the corresponding SINR derivations. In other cases RODS outperforms PANC_μ by roughly 2 dB. We conceive this to be a result of iterative convergence. The RODS algorithm can be extended to multi-user environment.

Appendix A
Relation Between Original and Distorted Data Symbols

We express $\tilde{a}_k$ given by (11) in vector form

$$\tilde{a}_k=a_k\cdot\alpha_1+\underline{J}^T\underline{\underline{L_k}}\underline{a}$$

$$\underline{a}_{N\times1}=\begin{pmatrix}a_0\\ \vdots\\ a_{N-1}\end{pmatrix};\quad \underline{J}_{N\times1}=\begin{pmatrix}\frac{1}{N}\sum_{d=2}^{D}\alpha_{2d-1}\gamma_0^{d-1}\\ \vdots\\ \frac{1}{N}\sum_{d=2}^{D}\alpha_{2d-1}\gamma_{N-1}^{d-1}\end{pmatrix};$$

$$\underline{\underline{L_k}}_{N\times N}=\begin{pmatrix}1 & \cdots & 1\\ e^{j2\pi\cdot1\cdot(-k)/N} & \cdots & e^{j2\pi\cdot1\cdot(N-k-1)/N}\\ e^{j2\pi\cdot2\cdot(-k)/N} & \cdots & e^{j2\pi\cdot2\cdot(N-k-1)/N}\\ \vdots & & \vdots\\ e^{j2\pi\cdot(N-1)\cdot(-k)/N} & \cdots & e^{j2\pi\cdot(N-1)\cdot(N-k-1)/N}\end{pmatrix};$$

$$\underline{\underline{L_k}}(n,p)=e^{j2\pi\cdot n\cdot(-k+p)/N}, \quad (21)$$

Authorized licensed use limited to: CAMBRIDGE UNIV. Downloaded on July 08,2026 at 03:40:52 UTC from IEEE Xplore. Restrictions apply.

where γ_n is the power of the signal $x(n)$, $\underline{a}$ is the complex data symbols vector, $\underline{\underline{L_k}}$ represents the interference caused by $\underline{J}$ and $\underline{a}$ to the kth subcarrier. The vectorial representation for all subcarriers is

$$\underline{\tilde{a}} = \left[\alpha_1 \underline{\underline{I}} + \begin{pmatrix} \underline{J}^T \underline{\underline{L_0}} \\ \vdots \\ \underline{J}^T \underline{\underline{L_{N-1}}} \end{pmatrix}\right] \underline{a} = \underline{\underline{K}} \cdot \underline{a}. \tag{22}$$

$\underline{\underline{I}}$ is the identity matrix. $\underline{J}^T \underline{\underline{L_k}}$ is of dimensions of $1 \times N$. $\underline{\underline{K}}$ is the matrix relating the received data symbols with the transmitted ones. The dependency in γ_n is unknown to the receiver, and is therefore estimated iteratively as given in Table II.

Appendix B
Derivation of the Nonlinear Compression Factor

Following (6) and observing only the elements in (11) that are dependent on a_k, we express μ as:

$$\mu = \alpha_1 + \frac{1}{N} \sum_{d=2}^{D} \sum_{n=0}^{N-1} \alpha_{2d-1} \gamma_n^{d-1}. \tag{23}$$

By the Central Limit Theorem, when N is sufficiently large, $x(n)$ can be assumed to be a complex Gaussian process with zero mean and variance $P_{av} = P/N$. Therefore $|x(n)| \sim$ Rayleigh$(2\sigma^2 = P/N)$ and

$$\begin{aligned} E(\gamma^{d-1}) &= \frac{1}{N} \sum_{n=0}^{N-1} \gamma_n^{d-1} = \frac{1}{N} \sum_{n=0}^{N-1} |x(n)|^{2(d-1)} \\ &= E(|x|^{2(d-1)}) = \sigma^{2(d-1)} (2d-3)!!. \end{aligned} \tag{24}$$

From (23) and (24) we can deduce μ up to the 15th order for the memoryless PA(1) as given in Table III:

$$\begin{aligned} \mu = \alpha_1 &+ \alpha_3 \cdot \frac{P}{N} + 2\alpha_5 \cdot \left(\frac{P}{N}\right)^2 + 6\alpha_7 \cdot \left(\frac{P}{N}\right)^3 \\ &+ 24\alpha_9 \cdot \left(\frac{P}{N}\right)^4 + 120\alpha_{11} \cdot \left(\frac{P}{N}\right)^5 + 720\alpha_{13} \cdot \left(\frac{P}{N}\right)^6 \\ &+ 5040\alpha_{15} \cdot \left(\frac{P}{N}\right)^7. \end{aligned} \tag{25}$$

Appendix C
Derivation of RODS Algorithm for Memory Polynomial

We extend the derivation of the RODS algorithm for the case of a power amplifier with memory, as given in (1). Substituting (1) into (4) yields

$$\tilde{a}_k = \sum_{n=0}^{N-1} \sum_{d=1}^{D} \sum_{q=0}^{Q} \alpha_{2d-1,q} x(n-\tau_q) |x(n-\tau_q)|^{2(d-1)} \cdot e^{-j2\pi k n/N} + n_0(k). \tag{26}$$

We define $\tilde{b_q}(k)$

$$\tilde{b_q}(k) = \sum_{n=0}^{N-1} \sum_{d=1}^{D} \alpha_{2d-1,q} x(n-\tau_q) |x(n-\tau_q)|^{2(d-1)} \cdot e^{-j2\pi k n/N}. \tag{27}$$

$\tilde{a}_k$ can thus be expressed as

$$\tilde{a}_k = \sum_{q=0}^{Q} \tilde{b_q}(k) + n_0(k). \tag{28}$$

Using the circular time shift property of OFDM we have

$$\mathrm{DFT}\{x(n-\tau_q)\} = \mathrm{DFT}\{x(n)\} \cdot e^{-j2\pi k \tau_q/N}. \tag{29}$$

By substituting (29) into (27) $\tilde{b_q}(k)$ can be expressed as

$$\tilde{b_q}(k) = \sum_{d=1}^{D} \alpha_{2d-1,q} \sum_{n=0}^{N-1} x(n) |x(n)|^{2(d-1)} \cdot e^{-j2\pi k n/N} e^{-j2\pi k \tau_q/N}. \tag{30}$$

By substituting $x(n)$ with their respective IDFT representation (3) into (30), we derive the following expression for the received subcarriers:

$$\begin{aligned} \tilde{b_q}(k) &= \alpha_{1,q} \cdot a_k \cdot e^{-j2\pi k \tau_q/N} \\ &\quad + \frac{1}{N} \sum_{d=2}^{D} \alpha_{2d-1,q} \cdot e^{-j2\pi k \tau_q/N} \\ &\quad \cdot \sum_{n=0}^{N-1} \sum_{p=0}^{N-1} a_p \gamma_n^{d-1} e^{j2\pi n(-k+p)/N} \\ &= a_k \mu_q(k) + \eta_q(k), \end{aligned} \tag{31}$$

where $\gamma_n = |x(n)|^2$ and

$$\begin{aligned} \mu_q(k) &= \alpha_{1,q} \cdot e^{-j2\pi k \tau_q/N} \\ &\quad + \frac{1}{N} \sum_{d=2}^{D} \alpha_{2d-1,q} \cdot e^{-j2\pi k \tau_q/N} \sum_{n=0}^{N-1} \gamma_n^{d-1}, \end{aligned} \tag{32}$$

$$\begin{aligned} \eta_q(k) &= \frac{1}{N} \sum_{d=2}^{D} \alpha_{2d-1,q} \cdot e^{-j2\pi k \tau_q/N} \\ &\quad \cdot \sum_{p=0, p\neq k}^{N-1} a_p \sum_{n=0}^{N-1} \gamma_n^{d-1} e^{j2\pi n(-k+p)/N}. \end{aligned} \tag{33}$$

The received subcarriers can be expressed as

$$\begin{aligned} \tilde{a_k} &= a_k \sum_{q=0}^{Q} \mu_q(k) + \sum_{q=0}^{Q} \eta_q(k) + n_0(k) \\ &= a_k \mu(k) + \eta(k) + n_0(k). \end{aligned} \tag{34}$$

For memory polynomial the complex coefficient μ is unique for a given subcarrier, hence dependent on the index k. We define $\alpha_{2d-1}(k)$ as

$$\alpha_{2d-1}(k) = \sum_{q=0}^{Q} \alpha_{2d-1,q} \cdot e^{-j2\pi k \tau_q/N}. \tag{35}$$

Substituting (31) and (35) into (28) we obtain

$$\begin{aligned} \tilde{a}_k &= \alpha_1(k) \cdot a_k \\ &\quad + \frac{1}{N} \sum_{d=2}^{D} \alpha_{2d-1}(k) \sum_{n=0}^{N-1} \sum_{p=0}^{N-1} a_p \gamma_n^{d-1} e^{j2\pi n(-k+p)/N} \\ &\quad + n_0(k), \end{aligned} \tag{36}$$

which resembles the memoryless case given in (11), but with the complex coefficient $\alpha_{2d-1}(k)$ unique for each subcarrier. The

Authorized licensed use limited to: CAMBRIDGE UNIV. Downloaded on July 08,2026 at 03:40:52 UTC from IEEE Xplore. Restrictions apply.

expression in (35) is used to derive the matrix $\underline{\underline{K}}$ for the memory polynomial case:

$$\underline{\underline{K}} = \left[\alpha_1 \underline{\underline{I}} + \begin{pmatrix} \underline{J_0}^T \underline{\underline{L_0}} \\ \vdots \\ \underline{J_{N-1}}^T \underline{\underline{L_{N-1}}} \end{pmatrix} \right], \tag{37}$$

where $\underline{J_k}$ is dependent on the subcarrier index k

$$\underline{J_k} = \begin{pmatrix} \frac{1}{N} \sum_{d=2}^{D} \alpha_{2d-1}(k) \gamma_0^{d-1} \\ \vdots \\ \frac{1}{N} \sum_{d=2}^{D} \alpha_{2d-1}(k) \gamma_{N-1}^{d-1} \end{pmatrix}. \tag{38}$$

Using the expressions in (37) and (38), the relation between the transmitted and received subcarriers is the same as given in (12), as is the RODS algorithm given in Table II.

Appendix D
Estimation of the Channel and Power Amplifier Model

OFDM-based standards typically employ a so-called *reference symbol*, i.e., an OFDM symbol consisting of pilot symbols that are used for channel estimation. Clearly, the OFDM reference symbol, may be corrupted when received due to power amplifier nonlinearity. It is challenging to accurately estimate the channel frequency response under such conditions. We distinguish between two cases: 1. the power amplifier model is known at the receiver side (e.g., the power amplifier is adequately modelled by a memory polynomial whose coefficients are extracted by the transmitter and conveyed to the receiver); 2. the power amplifier model coefficients are unknown and hence the receiver has to estimate both the model coefficients and the channel frequency response, needed for DPoD and channel estimation, respectively.

The preferable approach of the two above use cases should be chosen as follows. If the power amplifier is relatively static and the channel is noisy, or rapidly varying, the first approach would be more practical. For instance in a 4G home base station the power amplifier dynamics is such that a DPD learning mechanism is activated every few seconds or even minutes. At this relatively slow update rate the overhead in conveying the power amplifier model coefficients to the receiver would be small. Contrary, if the power amplifier model is dynamic and channel conditions allow it, the second approach may be more effective.

A. Scenario 1: Power Amplifier Model Is Known

We propose the following scheme for channel estimation given that the power amplifier model is known. Let $\underline{h}$ denote the channel impulse response, $\underline{x}$ an OFDM reference symbol and $\underline{y}$ the (nonlinearly) distorted (and amplified) version of $\underline{x}$ as defined by (1). The received reference symbol is given by

$$\underline{\tilde{y}} = \underline{y} * \underline{h} + \underline{n_0}, \tag{39}$$

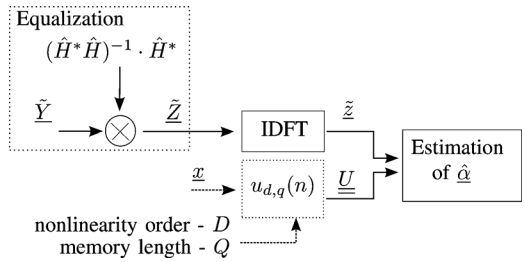


Fig. 9. power amplifier model estimation with (top) received predefined signal and (bottom) series $u_{dq}(n)$ composed of the predefined signal $\underline{x}$.

where $\underline{n_0}$ denotes an AWGN noise vector. In the frequency domain, this reads as

$$\underline{\tilde{a}} = \mathrm{DFT}(\underline{\tilde{y}}) = \underline{Y} \cdot \underline{H} + \mathrm{DFT}(\underline{n_0}). \tag{40}$$

Since both $\underline{x}$ and the power amplifier model are known at the receiver, so is $\underline{y}$, and the channel frequency response can be estimated as follows

$$\underline{\hat{H}} = (\underline{Y}^* \cdot \underline{Y})^{-1} \underline{Y}^* \cdot \underline{\tilde{a}}. \tag{41}$$

B. Scenario 2: Power Amplifier Model Is Unknown

When the power amplifier model is unknown, the receiver has to estimate the channel frequency response as well as the memory polynomial model coefficients. We first consider dividing this task into two separate processes: channel frequency response estimation, followed by power amplifier model estimation. Then, we consider combined estimation, that is estimation of a combined response of the power amplifier and the channel.

1) Separate Estimation of Channel and Power Amplifier Model: For accurate estimation of the channel frequency response, the distortion due to power amplifier nonlinearity has to be minimized. By minimizing the reference symbol PAPR, and setting adequate input back-off, the power amplifier may be assumed to be operating in the linear regime. Minimization of the PAPR can be achieved by choosing a subset of T out of the N available subcarriers, while the remaining $N-T$ subcarriers carry no data [18]. The number of subcarriers T to employ is closely related to the coherence BW of the channel. The 'pattern' of pilots must be denser than the channel coherence BW for sufficient frequency resolution, or otherwise channel estimation will fail.

DPoD techniques necessitate for the power amplifier model and the input back-off to be known at the receiver. The method for power amplifier model estimation should be chosen in accordance with desired estimation accuracy levels, as required by the chosen DPoD technique. power amplifier model estimation is analyzed in [18]. Fig. 9 presents a block diagram of the proposed method for power amplifier model estimation at the receiver side, where

- $\underline{x}$ denotes reference symbol.
- $\underline{\tilde{Y}}$ is the frequency domain representation of $\underline{y}$ as given by (1).
- $\underline{\hat{H}}$ is the estimated channel frequency response.
- $\underline{\tilde{Z}}$ is the equalizer output for $\underline{\tilde{Y}}$.
- $\underline{\tilde{z}}$ is the time domain representation of $\underline{\tilde{Z}}$.

Authorized licensed use limited to: CAMBRIDGE UNIV. Downloaded on July 08,2026 at 03:40:52 UTC from IEEE Xplore. Restrictions apply.

$$\underline{y}_{N\times 1} = \begin{pmatrix} y_0 \\ \vdots \\ y_{N-1} \end{pmatrix}; \quad \underline{\alpha}_{D(Q+1)\times 1} = \begin{pmatrix} \alpha_{1,0} \\ \vdots \\ \alpha_{D,0} \\ \vdots \\ \alpha_{1,Q} \\ \vdots \\ \alpha_{D,Q} \end{pmatrix}$$

$$\underline{\underline{U}}_{N\times D(Q+1)} = \begin{pmatrix} u_{10}(0) & \cdots & u_{D0}(0) & \cdots & u_{1Q}(0) & \cdots & u_{DQ}(0) \\ \vdots & & & & & & \vdots \\ u_{10}(N-1) & \cdots & u_{D0}(N-1) & \cdots & u_{1Q}(N-1) & \cdots & u_{DQ}(N-1) \end{pmatrix}$$

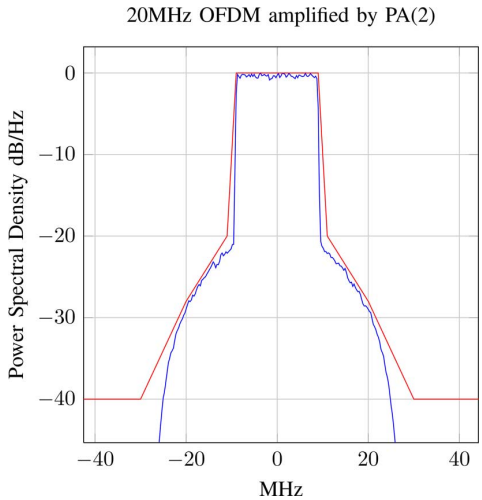


Fig. 10. Spectral emission of a 20 MHz OFDM signal amplified by PA(2) at input back-off of 8 dB. In red (top curve) is the 802.11ac spectral mask requirement.

We assume that the power amplifier has a nonlinearity order of D and memory taps $\tau_0, \ldots, \tau_Q$. Let us define the following series $u_{dq}(n)$, whose elements are:

$$u_{d,q}(n) = x(n-\tau_q)|x(n-\tau_q)|^{2(d-1)}. \tag{42}$$

We can represent the memory polynomial model in (1) in the following form:

$$\underline{y} = \underline{\underline{U}} \cdot \underline{\alpha}, \tag{43}$$

where (see the equation at the top of the page). Assuming perfect channel estimation, the equalized signal $\underline{\tilde{z}}$ equals the transmitted nonlinearly distorted signal $\underline{y}$ plus a colorful Gaussian noise. We propose to estimate the complex coefficients of the power amplifier model by the Least Squares solution for (43), based on the equalized received signal $\underline{\tilde{z}}$

$$\underline{\hat{\alpha}} = (\underline{\underline{U}}^* \cdot \underline{\underline{U}})^{-1} \cdot \underline{\underline{U}}^* \cdot \underline{\tilde{z}}. \tag{44}$$

The above estimation process should be repeated periodically to account for power amplifier changes, for example due to temperature variations.

2) Power Amplifier & Channel: Estimation of Combined Response: Estimation of the combined memory polynomial model is a subcase of the estimation method shown in Fig. 9, drawn by omitting the equalization block. The delay taps in (42) should represent the delay taps of the combined response. The estimated memory polynomial model is used by the RODS algorithm to perform joint DPoD and equalization to detect the received signal.

Appendix E
Spectral Emission

In Section III-B2 we discussed multi-user detection as an approach that could promote incorporation of DPoD in future wireless standards. In the following we demonstrate that post distortion can also benefit the current standards. Fig. 10 shows the spectrum of a 20 MHz OFDM signal amplified by PA(2) with back-off of 8 dB. The relevant 802.11ac spectral mask is also depicted. No predistortion is used, and the spectral mask is clearly not violated. For this case we have shown (Fig. 5) that the gain of DPoD over DPD is 2 dB.

Acknowledgment

The authors wish to thank the editor and the three anonymous referees for their helpful comments and suggestions.

References

[1] V. A. Bohara and S. H. Ting, "Analysis of OFDM signals in nonlinear high power amplifier with memory," in *Proc. IEEE Int. Conf. Commun.*, May 2008, pp. 3653–3657.
[2] V. A. Bohara and S. H. Ting, "Theoretical analysis of OFDM signals in nonlinear polynomial models," in *Proc. 6th Int. Conf. Inf., Commun. Signal Process.*, Dec. 2007, pp. 1–5.
[3] S. C. Cripps, *RF Power Amplifiers for Wireless Communications*, 2nd ed. Norwood, MA, USA: Artech House, 2006.
[4] G. T. Zhou, H. Qian, L. Ding, and R. Raich, "On the baseband representation of a bandpass nonlinearity," *IEEE Trans. Signal Process.*, vol. 53, no. 8, pp. 2953–2957, Aug. 2005.
[5] R. Raich and G. T. Zhou, "On the modeling of memory nonlinear effects of power amplifiers for communication applications," in *Proc. 10th IEEE Digit. Signal Process. Workshop*, Oct. 2002, pp. 1–6.
[6] H. Ku and J. S. Kenney, "Behavioral modeling of nonlinear RF power amplifiers considering memory effects," *IEEE Trans. Microw. Theory Tech.*, vol. 51, no. 12, pp. 2495–2504, Dec. 2003.

Authorized licensed use limited to: CAMBRIDGE UNIV. Downloaded on July 08,2026 at 03:40:52 UTC from IEEE Xplore. Restrictions apply.

[7] J. Reina-Tosina, C. Crespo-Cadenas, M. J. Madero-Ayora, and J. Munoz-Cruzado-Alba, "Performance of nonlinear behavioral models with OFDM signals," in *Proc. Integr. Nonlin. Microw. Millimetre-Wave Circuits*, Nov. 2008, pp. 105–108.
[8] J. Pan and C.-H. Cheng, "Electronic predistortion for compensation of fiber transmission impairments—Theory and complexity considerations," *J. Lightw. Technol.*, vol. 29, no. 18, Sep. 2011.
[9] Y. London and D. Sadot, "Nonlinear effects mitigation in coherent optical OFDM system in presence of high peak power," *J. Lightw. Technol.*, vol. 29, no. 21, Nov. 2011.
[10] L. Ding, G. T. Zhou, D. R. Morgan, Z. Ma, J. S. Kenney, J. Kim, and C. R. Giardina, "Memory polynomial predistorter based on the indirect learning architecture," in *Proc. IEEE Global Telecomm. Conf.*, Nov. 2002, pp. 967–971.
[11] J. Kim and K. Konstdntinou, "Digital predistortion of wideband signals based on power amplifier model with memory," *Electron. Lett.*, vol. 37, no. 23, pp. 1417–1418, Nov. 2001.
[12] L. Ding, G. T. Zhou, D. R. Morgan, Z. Ma, J. S. Kenney, J. Kim, and C. R. Giardina, "A robust digital baseband predistorter constructed using memory polynomials," *IEEE Trans. Commun.*, vol. 52, no. 1, pp. 159–165, Jan. 2004.
[13] C. Eun and E. J. Powers, "A new Volterra predistorter based on the indirect learning architecture," *IEEE Trans. Signal Process.*, vol. 45, no. 1, pp. 223–227, Jan. 1997.
[14] A. Zhu, P. J. Draxler, J. J. Yan, T. J. Brazil, D. F. Kimball, and P. M. Asbeck, "Open-loop digital predistorter for RF power amplifiers using dynamic deviation reduction-based Volterra series," *IEEE Trans. Microw. Theory Tech.*, vol. 56, no. 7, pp. 1524–1534, Jul. 2008.
[15] F. M. Ghannouchi and O. Hammi, "Behavioural modeling and predistortion," *IEEE Microw. Mag.*, vol. 10, no. 7, pp. 52–64, Dec. 2009.
[16] J. Tellado, L. M. C. Hoo, and J. M. Cioffi, "Maximum-likelihood detection of nonlinearly distorted multicarrier symbols by iterative decoding," *IEEE Trans. Commun.*, vol. 51, no. 2, Feb. 2003.
[17] J. Tellado, L. M. C. Hoo, and J. M. Cioffi, "Maximum-likelihood detection of nonlinearly distorted multicarrier symbols by iterative decoding," in *Proc. IEEE GlobeCom'99*, May 1999, pp. 2493–2498.
[18] F. H. Gregorio, S. Werner, J. Cousseau, J. Figueroa, and R. Wichman, "Receiver-side nonlinearities mitigation using an extended iterative decision-based technique," in *Proc. Signal Process.*, Mar. 2011, vol. 91, pp. 2042–2056.
[19] F. Gregorio, T. I. Laakso, and J. E. Cousseau, "Receiver cancellation of nonlinear power amplifier distortion in SDMA-OFDM systems," in *Proc. ICASSP*, 2006, vol. 4, pp. 325–328.
[20] A. Tehrani, H. Cao, A. Behravan, T. Eriksson, and C. Fager, "Successive cancellation of power amplifier distortion for multiuser detection," in *Proc. IEEE Veh. Technol. Conf.*, Sep. 2010, pp. 1–5.
[21] J. Gazda, P. Drotar, M. Deumal, D. Kocur, and P. Galajda, "Simple iterative cancellation of nonlinear distortion in LFDMA systems," in *Proc. 14th Int. OFDM Workshop*, 2009, pp. 133–137.
[22] M. Schetzen, *The Volterra and Wiener Theories of Nonlinear Systems*. New York, NY, USA: Wiley, 1980.
[23] J. Bussgang, Cross-Correlation Function of Amplitude-Distorted Gaussian Signals Res. Lab. Elect., Mass. Inst. Technol., Cambridge, MA, USA, Tech. Rep. 216, 1952.
[24] D. S. Papailiopoulos and G. N. Karystinos, "Near ml detection of nonlinearly distorted OFDM signals," in *Conf. Rec. 41st Asilomar Conf. Signals, Syst. Comp.*, Nov. 2007, pp. 1756–1760.
[25] H. Chen and A. M. Haimovich, "Iterative estimation and cancellation of clipping noise for OFDM signals," *IEEE Commun. Lett.*, vol. 7, no. 7, Jul. 2003.

Ziv Alina received the B.S. degree in electrical engineering and the B.A. degree in physics in 2006 and the M.S. degree in electrical engineering in 2015, all from Tel Aviv University, Israel.

He is currently with Qualcomm working as an algorithm engineer.

Ofer Amrani received the Ph.D. degree in electrical engineering from Tel-Aviv University, Tel-Aviv, Israel, in 2000.

In 2001, he joined the School of Electrical Engineering, Tel-Aviv University. His research interests are in the fields of communication, coding theory, and optical devices.

Authorized licensed use limited to: CAMBRIDGE UNIV. Downloaded on July 08,2026 at 03:40:52 UTC from IEEE Xplore. Restrictions apply.